

Week 6

Write a C program to simulate Real-Time CPU scheduling algorithms:

- Rate-Monotonic
- Earliest-deadline First

```
#include <stdio.h>
```

```
#include <math.h>
```

```
#define MAX 10
```

```
typedef struct d
```

```
int id;
```

```
int bt;
```

```
int dl;
```

```
int pr;
```

```
int ct;
```

```
int wt;
```

```
int tat;
```

```
} Process;
```

```
Process p[MAX];
```

```
int n;
```

```
float calculate_utilization() {
```

```
float U=0;
```

```
for (int i=0; i<n; i++) {
```

```
U += (float) p[i].bt / p[i].pr;
```

```
}
```

```
return U;
```

```
}
```

```
void cd
```



```
void cdf () {
```

```
    process temp [MAX];
```

```
    for (int i=0; i < n; i++)
```

```
        temp[i] = p[i];
```

```
    for (int i=0; i < n-1; i++)
```

```
        for (int j = i+1; j < n; j++)
```

```
            if (temp[i].de > temp[j].de)
```

```
                process t = temp[i];
```

```
                temp[i] = temp[j];
```

```
                temp[j] = t;
```

```
            }
```

```
        }
```

```
    }
```

```
    int time = 0;
```

```
    for (int i=0; i < n; i++)
```

```
        time += temp[i].bt;
```

```
        temp[i].ct = time;
```

```
        temp[i].tat = temp[i].ct;
```

```
        temp[i].wt = temp[i].tat - temp[i].bt;
```

```
    }
```

```
    float U = calculateUtilization();
```

```
    printf("\n Earliest Deadline First (EDF) Scheduling == (n)");
```

```
    printf("Utilization (U) = %f\n", U);
```

```
    if (U <= 1)
```

```
        printf("Schedulable (U <= 1)\n");
```

```
    else
```

```
        printf("Not schedulable (U > 1)\n");
```

```
    printf("\n ID | BT | CT | ETAT | WT | TAT | (n)");
```

```
    for (int i=0; i < n; i++)
```

```
        printf("%d | %d | %d | %d | %d | %d | \n",
```

```
            temp[i].id, temp[i].bt, temp[i].ct,
```

```
            temp[i].et, temp[i].wt, temp[i].tat);
```

```
    }
```

```
}
```



```
void rms () {
```

```
    Process kmp [MAX];
```

```
    for (int i=0; i<n; i++)
```

```
        kmp[i] = p[i];
```

```
    for (int i=0; i<n-1; i++)
```

```
        for (int j=i+1; j<n; j++)
```

```
            if (kmp[i].pr > kmp[j].pr)
```

```
                Process t = kmp[i];
```

```
                kmp[i] = kmp[j];
```

```
                kmp[j] = t;
```

```
        }
```

```
    }
```

```
    int time = 0;
```

```
    for (int i=0; i<n; i++)
```

```
        time += kmp[i].bt;
```

```
        kmp[i].ct = time;
```

```
        kmp[i].tat = kmp[i].ct;
```

```
        kmp[i].wt = kmp[i].tat - kmp[i].bt;
```

```
    }
```

```
    float U = calculate_utilization (n);
```

```
    float bound = n * (power(2, (float)1/n) - 1);
```

```
    printf ("Rate Monotonic Scheduling (RMS) is feasible\n");
```

```
    printf ("Utilization (U) = %.3f\n", U);
```

```
    printf ("RM Bound = %.3f\n", bound);
```

```
    if (U <= bound)
```

```
        printf ("Schedulable (U <= Bound)\n");
```

```
    else
```

```
        printf ("Not schedulable (U > Bound)\n");
```

```
    for (int i=0; i<n; i++)
```

```
        printf ("%d\t%d\t%d\t%d\t%d\t%d\n",
```

```
            kmp[i].id, kmp[i].bt, kmp[i].pr,
```

```
            kmp[i].ct, kmp[i].wt, kmp[i].tat);
```



```

int main() {
    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("\n Enter process details: \n");
    for(int i=0; i<n; i++) {
        p[i].id = i;
        printf("\n Process %d: \n", i);
        printf("Burst Time: ");
        scanf("%d", &p[i].bt);
        printf("Deadline (for EDF): ");
        scanf("%d", &p[i].dl);
        printf("Period (for RMS): ");
        scanf("%d", &p[i].pr);
    }

    edfc();
    rms();
    return 0;
}

```

O/P:

Enter number of processes: 3

Enter process details:

Process 0:

Burst Time: 3

Deadline (for EDF): 9

Period (for RMS): 520

Process 1:

Burst time: 2

Deadline (for EDF): 8

Period (for RMS): 10

==== Earliest Deadline First (EDF) Scheduling ====

Utilization (U) = 0.750

Schedulable (U ≤ 1)

ID	BT	Deadline	CT	WT	TAT
1	2	4	2	0	2
0	3	7	5	2	5
2	2	8	7	5	7

==== Rate Monotonic Scheduling (RMS) ====

Utilization (U) = 0.750

RMS Bound = 0.780

Schedulable (U ≤ Bound)

ID	BT	Period	CT	WT	TAT
1	2	5	2	0	2
2	2	10	4	2	4
0	3	20	7	4	7

Enter number of processes: 3

Enter process details:

Process 0:

Burst time: 2

Deadline (for EDF): 5

Period (for RMS): 5

Process 1:

Burst time: 2

Deadline (for EDF): 7

Period (for RMS): 2

==== Earliest Deadline First (EDF) scheduling =====

Utilization (U) = 1.061

Not Schedulable ($U > 1$)

ID	BT	Deadline	CT	WT	TAT
0	2	5	2	0	2
1	2	7	4	2	4
2	3	8	7	4	7

3/1/2